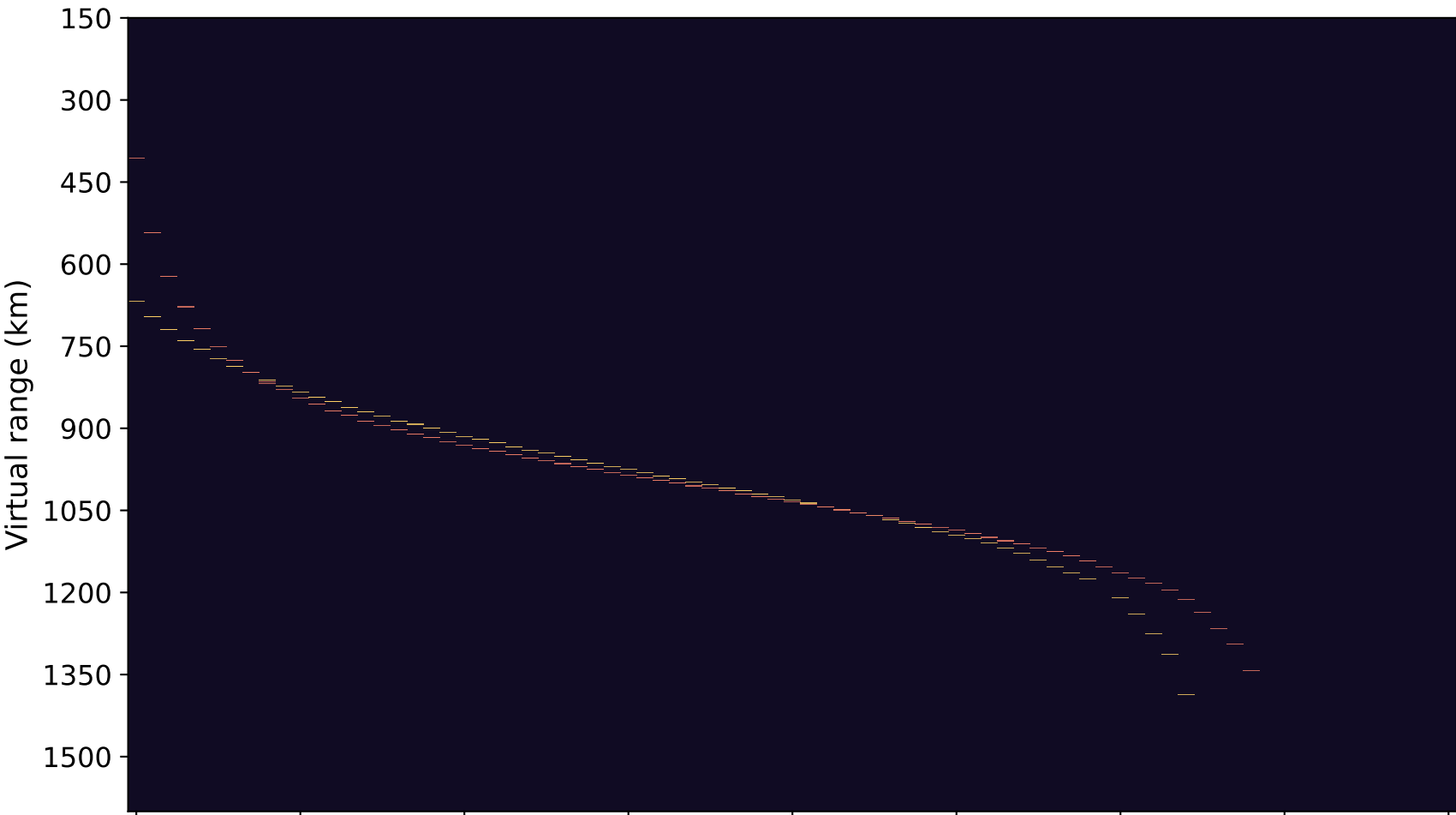


3D-wave ionograms

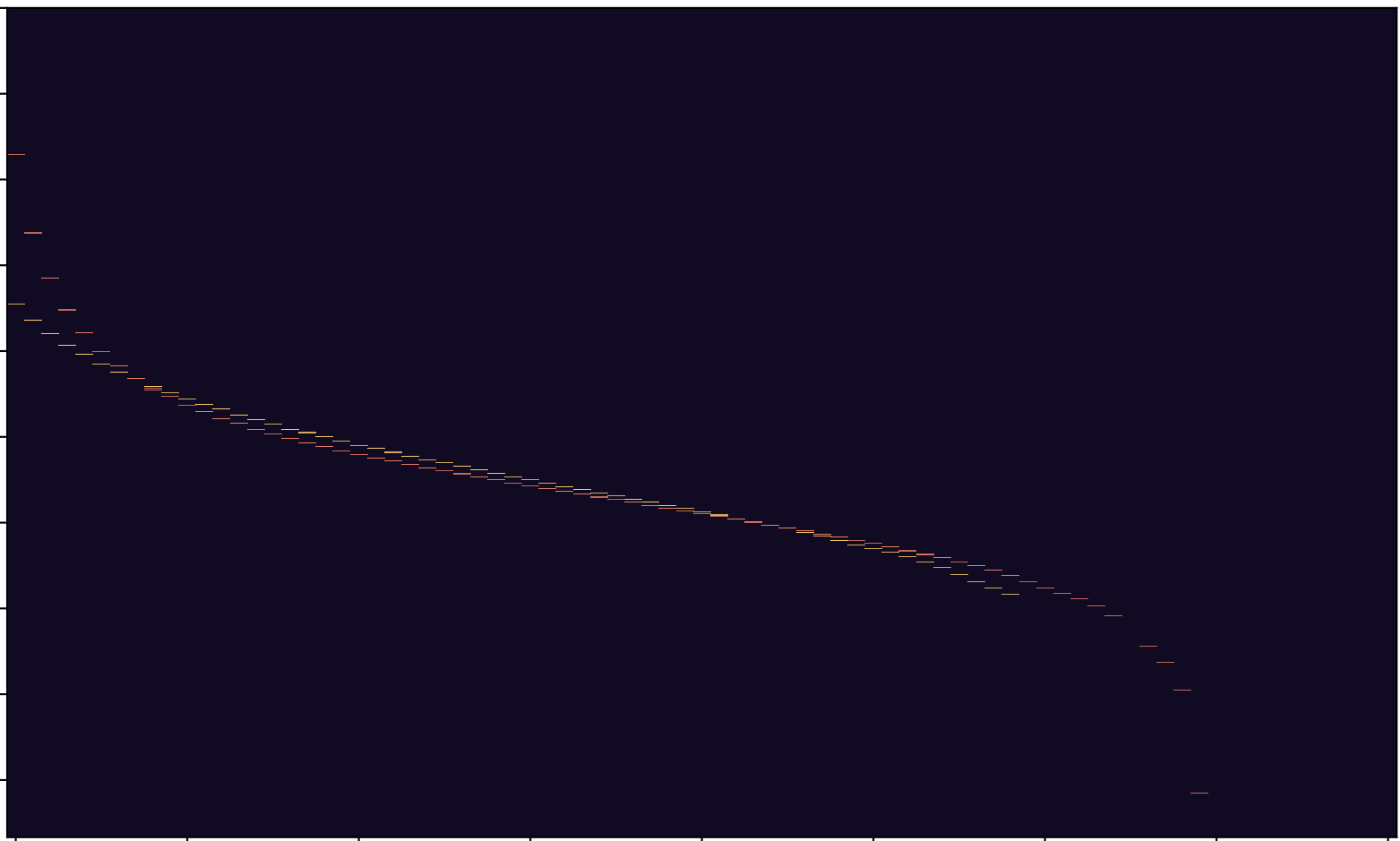
Synthetic ionosphere · 2–10 MHz in 100 kHz steps · 1 km range bins · all accepted O/X returns

O mode X mode

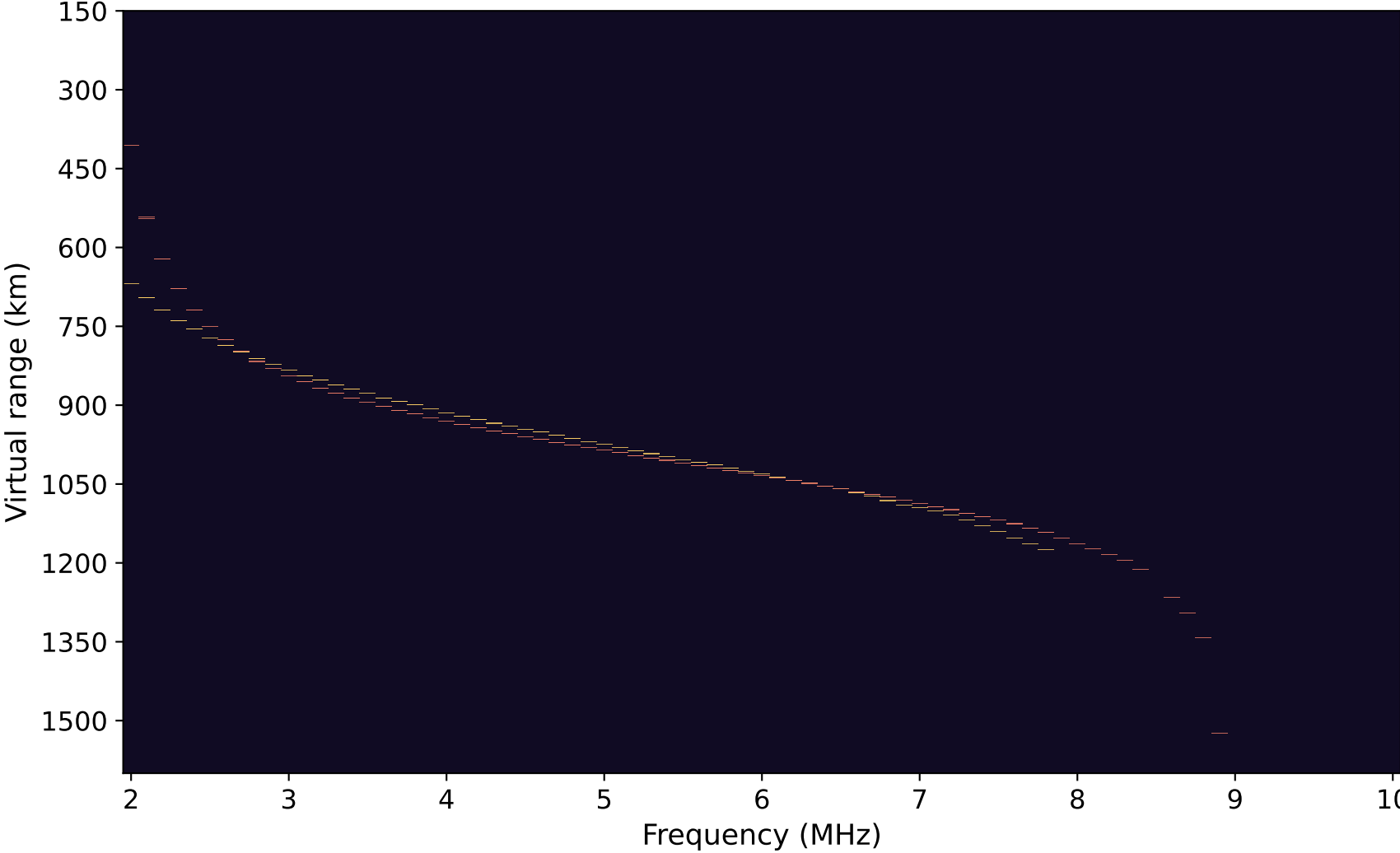
A Original az/el · 215 returns · 75.8 s
168 anchor / 288 full directions · reference



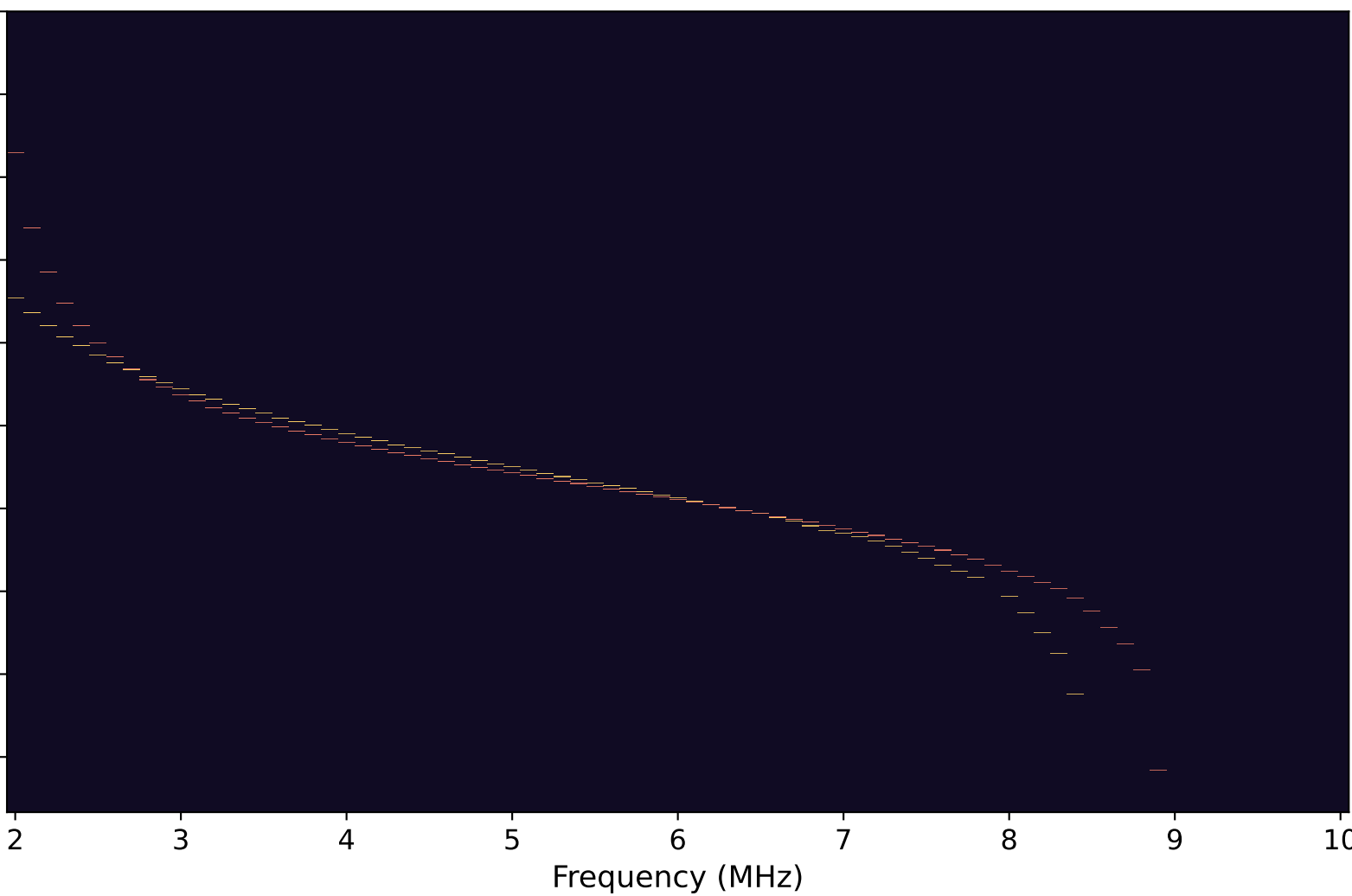
B Guarded full outer fan, 0 extra seeds · 211 returns · 48.0 s
174 anchor / 288 full directions · 208/215 reference returns matched



C Guarded half outer fan, 0 extra seeds · 211 returns · 45.1 s
105 anchor / 162 full directions · 208/215 reference returns matched



D Guarded half outer fan, 4 extra seeds · 216 returns · 70.2 s
105 anchor / 162 full directions · 215/215 reference returns matched



Homing gate: 1,000 m throughout. Match: same frequency and mode, virtual ranges within 1 km; coincident returns may blend. Timings cover both tracing modes; shared grid setup excluded.